Lab 4: Make a signal generator

To test medical devices, we often use fake signals. We can do this measure the accuracy if we can generate precise known signals. We can use it to verify that a medical device is still functioning, which is a task often done at the MT-department in a hospital. We can use it during development to test our prototype. The last item is the target of this lab. We will use the signal generator to test the prototype that we developed in Lab 1: The PPG-based heart rate device!

The signal generator must be able to do the following:

1. Generate an electrical signal in the range between 0-3V. The wave form should resemble a typical PPG-based curve.
2. The user must be able to choose different heart rate values for the generated curve. The selection is done with the buttons. The lowest must be 30 BPM and the highest be 150 BPM. The same wave form can be used for all heart rates.
3. The accuracy of the generated heart rate should not deviate more than ±5% from the selected heart rate.
4. The signal generator is built with an Micro:Bit and analogue electronics soldered to a protoboard. The signal output should be provided through a pin-header so that common jumper wires can be used to connect the signal generator with the Lab 1 device.

It is advised to first try out your analogue electronics on a breadboard and check the output with either an oscilloscope or the Lab 1 device.

The Micro:Bit does not have a proper Digital-to-Analogue Converter (DAC). Therefore, you need to build one yourself. The Micro:Bit only provides digital outputs, but the digital outputs can be toggled between 0V and 3V in a very rapid fashion using a technique called pulse width modulation (PWM). You need to read up on this topic and how to do this on the Micro:Bit. Furthermore, do not use MakeCode to do this. Instead use Python: https://python.microbit.org/

https://microbit-micropython.readthedocs.io/en/v2-docs/pin.html

The PWM signal does not behave as a heart signal. However, if we reduce the PWM period time, and pass the signal through a low-pass filter, we can generate a quite stable voltage value for a short period of time. Then we can keep that voltage for a few milliseconds and the change it. By changing the level with regular intervals, we can generate the PPG wave form.

However, you need to design an analogue low-pass filter. Here is a web page that will instruct you how you can do this. Start simple and see if it works. If you are not happy with the simplest design, continue to the next simplest design and try that one. Repeat until you are happy.

https://www.electronics-tutorials.ws/filter/filter_2.html

Passive components and other small cheap stuff (e.g., pin headers, jumper wires, etc.) can be found in MakerSpace. If you need op-amps, we can provide you with such too. We have Microchip MCP6002 in PDIP-package.

When you are happy with the filter, solder the components onto the protoboard.
Here is good video on how to solder protoboards:
https://www.youtube.com/watch?v=l9Kbr8cPqOE

Examination

When ready with the lab, you need to orally demonstrate your solution. You will book a time slot for this. Be prepared to show your work with both your lab 1 device and on the oscilloscope. If done remotely, make a recording of your oscilloscope test in advance and show the recording to the teacher.

Important Notes about connecting the Micro:Bit to the protoboard

NEVER solder the Micro:Bit!!!

Also **do not solder the Micro:bit adaptor**.

Use the cables to connect to the protoboard. It is okay to cut and/or solder the cables!

For instance, connect a jumper wire to the leg of the adaptor directly or via a breadboard. The jumper wire can then be connected to the protoboard by cutting it and soldering it directly, or via pin-header soldered to the protoboard.

Some hints

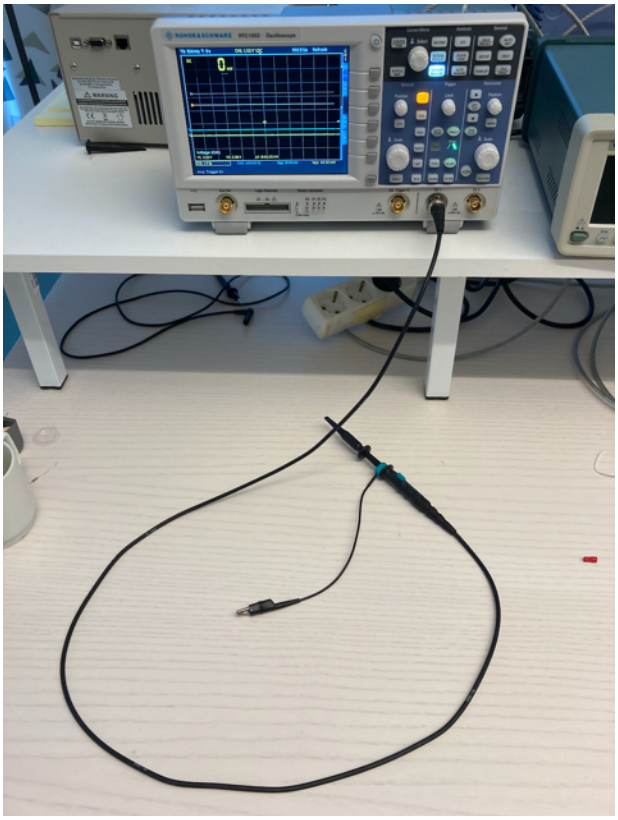
What resistance R and capacitance C to choose?
http://www.doctormonk.com/2020/08/testing-microbits-analog-inputs.html
Make sure the capacitor C is to be found in MakerSpace. The larger the capacitance, the more expensive they are and the bigger they are. Hence, generally we do not stock capacitors above ~ 10 uF. (For SMD capacitors and electrolyte capacitors, we have larger capacitance, but you do not need them)

What cut-off frequency to choose?
What frequency does your PWM signal have? Calculate it! Make sure it becomes nearly a flat line when you do not change the duty cycle/period.

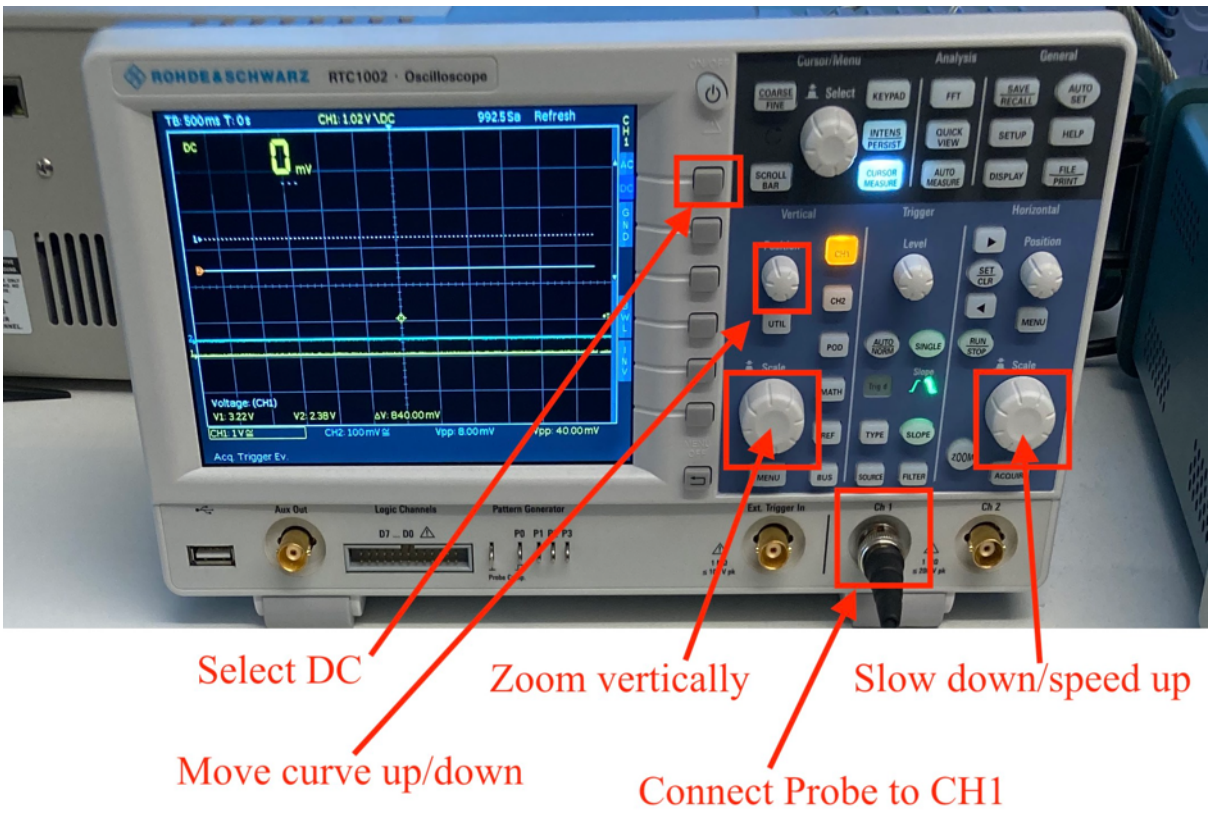
However, do not filter out the frequencies of the PPG-signal. So what are the highest frequency components of an PPG-signal? Yes, it is hard to know unless you have experience with DFT (Discrete Fourier Transform). So another approach is to try a filter and watch the wave form on the oscilloscope and determine if the wave form is altered too much.

Where can I find an PPG curve?
The MIMIC-3 waveform database will have some curve data for you. You are allowed take ECG or ABP wave forms instead if you want. PPG is called PLETH.
https://archive.physionet.org/physiobank/database/mimic3wdb/
https://archive.physionet.org/cgi-bin/atm/ATM?database=mimic3wdb&tool=plot_waveforms

How do I use the oscilloscope?
First, use our more modern oscilloscopes by Rodhe & Schwartz RTC1002. You find the manual here: https://www.rohde-schwarz.com/webhelp/RTC_HTML_UserManual_en/Content/welcome.htm



Connect the probe to CH1 and attach the crocodile clip to the ground on your circuit or the Micro:Bit ground. Then hook of the probe should be connected to the signal cable.



If the signal does not show up on the screen, do the following:

- Make sure DC is selected for CH1
- Turn off triggering. You should get the signal continuously
- Turn the vertical position and scale nobs to move the curve up/down or “zoom in/out”
- Turn the horizontal scale nob to slow down speed up the signal